Student: _____ **Date:** _____	**Instructor:** Abel Embaye **Course:** ECON 31303-901/2 - INTERMEDIATE MACROECONOMICS (fall2026)	**Assignment:** Sample Exam 2 (Ch 6, 7, 8 & 9)

1. The difference between the growth rate of labor productivity and the growth rate of total factor productivity depends on

   - **A.** the growth rate of the amount of capital per unit of labor.
   - **B.** the growth rate of real wages.
   - **C.** the growth rate of business investment.
   - **D.** the inflation rate.

   From 1995 to 2008, the growth rate of capital per unit of labor was high, which caused

   - **A.** high inflation rates.
   - **B.** labor productivity growth to be much higher than total factor productivity growth.
   - **C.** labor productivity growth to be much lower than total factor productivity growth.
   - **D.** wages to grow faster than before.

   Answers: A. the growth rate of the amount of capital per unit of labor.
   B. labor productivity growth to be much higher than total factor productivity growth.

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2. In the Solow model, the saving rate (*s*) is

   - **A.** unknown.
   - **B.** exogenous.
   - **C.** endogenous.
   - **D.** a growth rate.

   In the Solow model, steady-state consumption per worker (*c*) is

   - **A.** exogenous.
   - **B.** unknown.
   - **C.** endogenous.
   - **D.** a growth rate.

   In the Solow model, a sensible question to ask is

   - **A.** how does a change in steady-state consumption per worker (*c*) affect the saving rate (*s*)?
   - **B.** how does a change in steady-state consumption per worker (*c*) affect the steady-state population growth rate (*n*)?
   - **C.** how does a change in the saving rate (*s*) affect the steady-state population growth rate (*n*)?
   - **D.** how does a change in the saving rate (*s*) affect steady-state consumption per worker (*c*)?

   Answers: B. exogenous.
   C. endogenous.
   D. how does a change in the saving rate (*s*) affect steady-state consumption per worker (*c*)?

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3. According to the Solow model of economic growth, if there is no productivity growth, what will happen to output per worker, consumption per worker, and capital per worker in the long run?

   - **A.** all will increase
   - **B.** all will decrease
   - **C.** all will remain the same
   - **D.** output and capital per worker will increase, while consumption will decrease

   Answer: C. all will remain the same

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4. The higher the steady-state capital-labor ratio is, the more consumption each worker can enjoy in the long run.

   True or false?

   - **A.** True

   Answer: B. False

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5. What two explanations of productivity growth does endogenous growth theory offer?

   - **A.** population growth and increased saving
   - **B.** population growth and technological innovation
   - **C.** increased saving and technological innovation
   - **D.** accumulation of human capital and technological innovation

   How does the production function in an endogenous growth model differ from the production function in the Solow model? In the Solow model, the production function _____, while in the endogenous growth model, the production function _____.

   - **A.** depends on capital but not labor; depends on labor but not capital
   - **B.** depends on labor but not capital; depends on capital but not labor
   - **C.** does not exhibit diminishing marginal productivity; exhibits diminishing marginal productivity
   - **D.** exhibits diminishing marginal productivity; does not exhibit diminishing marginal productivity.

   Answers: D. accumulation of human capital and technological innovation
   D. exhibits diminishing marginal productivity; does not exhibit diminishing marginal productivity.

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6. Some economists argue that the U.S. saving rate is too low. Which of the following is a reasonable explanation for why the invisible hand of the market has failed to achieve a saving rate which optimally balances present consumption against future increases in living standards?

   - **A.** Americans are predisposed to overweight the value of present consumption as compared to future consumption.
   - **B.** Tax treatment of income discourages saving by taxing the returns to saving.
   - **C.** The political process results in deficit spending that differs from public preferences.
   - **D.** All of these are reasonable explanations.
   - **E.** None of these are reasonable explanations.

   Answer: D. All of these are reasonable explanations.

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7. How does the *expected rate of return* on an asset affect its desirability?

   - **A.** They are inversely related - the higher the expected return, all else equal, the less attractive the asset.
   - **B.** They are directly related - the higher the expected return, all else equal, the more attractive the asset.
   - **C.** They are unrelated - expected return does not affect an asset's desirability.

   How does the level of *risk* affect an asset's desirability?

   - **A.** They are inversely related - the higher the level of risk, all else equal, the less attractive the asset.
   - **B.** They are directly related - the higher the level of risk, all else equal, the more attractive the asset.
   - **C.** They are unrelated - an asset's level of risk is unrelated to its desirability.

   How does the level of *liquidity* affect an asset's desirability?

   - **A.** They are inversely related - the more liquid the asset, all else equal, the less desirable the asset.
   - **B.** They are directly related - the more liquid the asset, all else equal, the more desirable the asset.
   - **C.** They are unrelated - liquidity does not affect an asset's desirability.

   Answers: B. They are directly related - the higher the expected return, all else equal, the more attractive the asset.
   A. They are inversely related - the higher the level of risk, all else equal, the less attractive the asset.
   B. They are directly related - the more liquid the asset, all else equal, the more desirable the asset.

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8. Define the velocity of money.

- **A.** It is the ratio of the amount of money in circulation to the price level.
- **B.** It is the ratio of real GDP to the amount of money in circulation.
- **C.** It is the ratio of nominal GDP to the price level.
- **D.** It is the ratio of nominal GDP to the amount of money in circulation.

In the quantity theory of money, velocity is assumed to be highly (1) _____.

(1)
- volatile
- stable

Answers: D. It is the ratio of nominal GDP to the amount of money in circulation.
(1) stable

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9. Which of the following four variables has little impact on the aggregate demand for money?

- **A.** the real exchange rate
- **B.** the price level
- **C.** aggregate income
- **D.** the nominal interest rate

Answer: A. the real exchange rate

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10. Consider the equilibrium condition for the asset market $\frac{M}{P} = L\left(Y, r + \pi^e\right)$.

Using this condition, answer the following questions.

A decrease in inflation expectations(π^e), with no change in output, real interest rate, or the money supply, will result in

(1) _____ in the price level.

An increase in the nominal money supply (M), with no change in output, real interest rate, or inflation expectations, will result in

(2) _____ in the price level.

A decrease in output (Y), with no change in the money supply, real interest rate, or inflation expectations, will result in

(3) _____ in the price level.

(1)	(2)	(3)
an increase	an increase	an increase
no change	no change	no change
a decrease	a decrease	a decrease

Answers: (1) a decrease
(2) an increase
(3) an increase

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11. Which of the following is most likely to lead to a decrease of 10% in the nominal demand for money?

- **A.** A decrease in real income of 5%.
- **B.** An increase of 10% in the price level.
- **C.** A decline of 10% in the price level.
- **D.** An increase in real income of 5%.

Answer: C. A decline of 10% in the price level.

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12. Which of these variables is not a variable in the equation for the asset market equilibrium condition?

- **A.** Real income
- **B.** Expected rate of inflation
- **C.** Real interest rate
- **D.** Saving

Answer: D. Saving

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13. Which of the following is NOT a reasonable explanation for the Great Moderation?

- **A.** higher oil prices
- **B.** changes in the behavior of the labor market
- **C.** changes in technology
- **D.** improved supply-chain management



Answer: A. higher oil prices

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14. The deepest contraction in U.S. history occurred

- **A.** in the 1960s.
- **B.** from 2007 to 2009.
- **C.** during the 1930s.
- **D.** in the 1970s.

Since World War II, the longest U.S. recession occurred

- **A.** in the 1990s.
- **B.** in the 1980s.
- **C.** in the 1970s.
- **D.** in the 2000s.

Answers: C. during the 1930s.
D. in the 2000s.

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15. What terms are used to describe the way a variable moves in terms of direction when economic activity is rising or falling?

- **A.** procyclical and countercyclical
- **B.** peak and trough
- **C.** stable and unstable
- **D.** leading, coincident, and lagging

What terms are used to describe the timing of cyclical changes in economic variables?

- **A.** peak and trough
- **B.** stable and unstable
- **C.** procyclical and countercyclical
- **D.** leading, coincident, and lagging

Answers: A. procyclical and countercyclical
D. leading, coincident, and lagging

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16. Identify the following variables as being procyclical or countercyclical:

Unemployment: (1) _____

Consumption Expenditure: (2) _____

Inflation: (3) _____

Nominal Interest Rates: (4) _____

(1) ○ Procyclical ○ Countercyclical
(2) ○ Procyclical ○ Countercyclical
(3) ○ Procyclical ○ Countercyclical
(4) ○ Procyclical ○ Countercyclical

Answers: (1) Countercyclical
(2) Procyclical
(3) Procyclical
(4) Procyclical

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17. If you knew that the economy was falling into a recession, what would you expect to happen to production during the next few quarters?

(1) _____

To investment? (2) _____

To average labor productivity? (3) _____

To the real wage? (4) _____

To the unemployment rate? (5) _____

(1) ○ Increase ○ Decrease ○ Remain unchanged
(2) ○ Increase ○ Decrease ○ Remain unchanged
(3) ○ Increase ○ Decrease ○ Remain unchanged
(4) ○ Increase ○ Decrease ○ Remain unchanged

(5) ○ Increase ○ Decrease ○ Remain unchanged

Answers: (1) Decrease
(2) Decrease
(3) Decrease
(4) Decrease
(5) Increase

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18. Consumer expenditures on durable goods such as cars and furniture, as well as purchases of new houses, fall much more than expenditures on nondurable goods and services during most recessions. Why do you think that is?

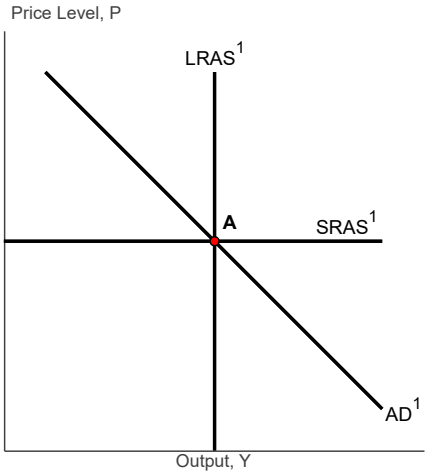
○ **A.** because companies stop making cars, furniture, and houses in recessions
○ **B.** because people can more easily change the timing of their expenditure on durables
○ **C.** because prices of cars, furniture, and houses rise sharply in recessions
○ **D.** because the government increase taxes on cars, furniture, and houses in recessions

Answer: B. because people can more easily change the timing of their expenditure on durables

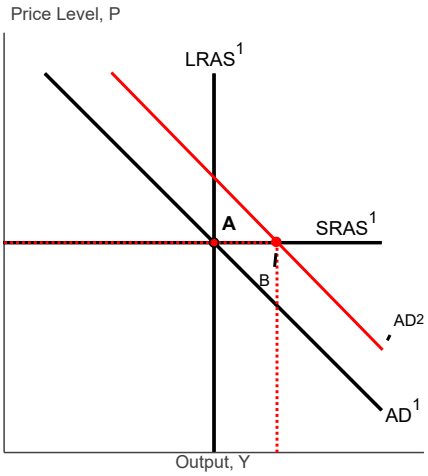
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19. 1. *Use the line drawing tool* to show the effects of a positive aggregate demand shock that might result from consumers becoming more optimistic. Properly label this line.
2. *Use the point drawing tool* to identify the new short run equilibrium. Label this point 'B'.

Carefully follow the instructions above, and only draw the required objects.



Answer:



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20. How do Keynesians and classicals differ in their beliefs about how long it takes the economy to reach long-run equilibrium? What implications do these differences in beliefs have for Keynesian and classical views about the usefulness of antirecessionary policies? Classical economists think prices adjust _____ and that antirecessionary policies are _____, whereas Keynesian economists think the opposite.

- **A.** rapidly; necessary
- **B.** slowly; not necessary
- **C.** rapidly; not necessary
- **D.** slowly; necessary

What implications do these differences in beliefs have for the types of shocks that cause most recessions? Classical economists think that _____ shocks cause recessions, whereas Keynesian economists think that _____ shocks cause recessions.

- **A.** both aggregate demand and aggregate supply; aggregate demand
- **B.** aggregate supply; both aggregate demand and aggregate supply
- **C.** both aggregate demand and aggregate supply; aggregate supply
- **D.** aggregate demand; both aggregate demand and aggregate supply

Answers: C. rapidly; not necessary
B. aggregate supply; both aggregate demand and aggregate supply

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21. The position of the *FE* line is determined by:

- **A.** the price level and interest rate.
- **B.** the labor market and the production function.
- **C.** the money supply and government purchases.
- **D.** consumption and investment.

All of the following will shift the *FE* line except:

- **A.** a beneficial supply shock.
- **B.** an increase in the price level.
- **C.** an increase in labor supply.
- **D.** an increase in the capital stock.

Answers: B. the labor market and the production function.
B. an increase in the price level.

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22. The *IS* curve represents:

- **A.** combinations of output and the real interest rate such that desired national saving is equal to desired investment.
- **B.** combinations of output and the real interest rate such the output is equal to full-employment output.
- **C.** combinations of output and the real interest rate such that the real interest rate is equal to the full-employment interest rate.
- **D.** combinations of output and the real interest rate such that money demand is equal to money supply.

Answer: A. combinations of output and the real interest rate such that desired national saving is equal to desired investment.

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23. In each of the following cases, what is the effect on the *IS* curve?

An increase in the effective tax rate on capital

- **A.** shifts the *IS* curve down and to the left.
- **B.** shifts the *IS* curve up and to the right.
- **C.** does not change the *IS* curve.

An increase in the money supply

- **A.** shifts the *IS* curve up and to the right.
- **B.** does not change the *IS* curve.
- **C.** shifts the *IS* curve down and to the left.

A temporary increase in goverment spending

- **A.** does not change the *IS* curve.
- **B.** shifts the *IS* curve up and to the right.
- **C.** shifts the *IS* curve down and to the left.

Answers: A. shifts the *IS* curve down and to the left.
B. does not change the *IS* curve.
B. shifts the *IS* curve up and to the right.

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24. In each of the following cases, what is the effect on the *LM* curve?

An increase in the expected inflation rate

- ○ **A.** shifts the *LM* curve up and to the left.
- ○ **B.** shifts the *LM* curve down and to the right.
- ○ **C.** does not shift the *LM* curve.

An increase in government spending

- ○ **A.** shifts the *LM* curve up and to the left.
- ○ **B.** shifts the *LM* curve down and to the right.
- ○ **C.** does not shift the *LM* curve.

An increase in the price level

- ○ **A.** shifts the *LM* curve up and to the left.
- ○ **B.** does not shift the *LM* curve.
- ○ **C.** shifts the *LM* curve down and to the right.

Answers: B. shifts the *LM* curve down and to the right.
C. does not shift the *LM* curve.
A. shifts the *LM* curve up and to the left.

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25. Consider the general-equilibrium effects of a temporary adverse supply shock.

For each of the following variables, select whether you would expect it to increase, decrease, or remain constant in general equilibrium as a result of the temporary supply shock.

Output (1) _____

The real interest rate (2) _____

The price level (3) _____

For each of the following variables, select whether you would expect it to increase, decrease, or remain constant as a result of the temporary adverse supply shock.

Consumption (4) _____

Investment (5) _____

Government spending (6) _____

The nominal money supply (7) _____

(1) ○ will increase. ○ will stay the same. ○ will decrease.

(2) ○ will increase. ○ will stay the same. ○ will decrease.

(3) ○ will increase. ○ will stay the same. ○ will decrease.

(4) ○ will increase. ○ will stay the same. ○ will decrease.

(5) ○ will increase. ○ will stay the same. ○ will decrease.

(6) ○ will increase. ○ will stay the same. ○ will decrease.

(7) ○ will increase. ○ will stay the same. ○ will decrease.

Answers: (1) will decrease.
(2) will increase.
(3) will increase.
(4) will decrease.
(5) will decrease.
(6) will stay the same.
(7) will stay the same.

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26. Use the *IS-LM* model to determine the effects of an increased usage of automatic teller machines that reduces thedemand for money. Show the effects below on the general equilibrium values of real wage, employment, output, real interest rate, consumption, investment, and price level.

Real wage (1) _____ Employment (2) _____

Output (3) _____ Price Level (4) _____

Real Interest Rate (5) _____ Investment (6) _____

Consumption (7) _____

(1) ○ Remains unchanged ○ Increases ○ Decreases

(2) ○ Remains unchanged ○ Increases ○ Decreases

(3) ○ Remains unchanged ○ Decreases ○ Increases

(4) ○ Decreases ○ Increases ○ Remains unchanged

(5) ○ Increases ○ Remains unchanged ○ Decreases

(6) ○ Increases ○ Decreases ○ Remains unchanged

(7) ○ Increases ○ Decreases ○ Remains unchanged

Answers: (1) Remains unchanged
(2) Remains unchanged
(3) Remains unchanged
(4) Increases
(5) Remains unchanged
(6) Remains unchanged
(7) Remains unchanged

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27. When economists say that money is neutral, this means that:

○ **A.** a change in the money supply changes real variables but not nominal variables.
○ **B.** a change in the money supply has no effect on the economy.
○ **C.** a change in the money supply changes nominal variables but not real variables.
○ **D.** a change in the money supply will stall the economy, preventing further growth.

Answer: C. a change in the money supply changes nominal variables but not real variables.

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28. In the *AD-AS* model, the short-run effect of a decrease in the money supply is

○ **A.** a shift down and to the left of the *AD* curve, causing output to fall at an unchanged price level.
○ **B.** no change, because money is neutral in the short run.
○ **C.** a shift up and to the right of the *AD* curve, causing output to rise at an unchanged price level.
○ **D.** a shift up and to the right of the *AD* curve and a shift up of the *SRAS* curve, causing output and the price level to rise.
○ **E.** a shift down and to the left of the *AD* curve and a shift down of the *SRAS* curve, causing output and the price level to fall.

In the *AD-AS* model, the long-run effect of a decrease in the money supply is

○ **A.** a proportionate fall in the price level and in output.
○ **B.** no change at all, because money is neutral in the long run.
○ **C.** a proportionate increase in the price level, but no changes to real variables such as output.
○ **D.** a proportionate increase in the price level and in output.
○ **E.** a proportionate fall in the price level, but no changes to real variables such as output.

Answers: A. a shift down and to the left of the *AD* curve, causing output to fall at an unchanged price level.
E. a proportionate fall in the price level, but no changes to real variables such as output.

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29. Describe the short-run aggregate supply (*SRAS*) curve and the long-run aggregate supply (*LRAS*) curve.

- **A.** the *SRAS* curve is horizontal and the *LRAS* curve is upward sloping
- **B.** the *SRAS* curve is vertical and the *LRAS* curve is upward sloping
- **C.** the *SRAS* curve is horizontal and the *LRAS* curve is vertical
- **D.** the *SRAS* curve is vertical and the *LRAS* curve is horizontal

Why is the short-run aggregate supply curve horizontal?

- **A.** because prices remain fixed in the short run
- **B.** because the real interest rate is fixed in the short run
- **C.** because capital is fixed in the short run
- **D.** because output is fixed in the short run

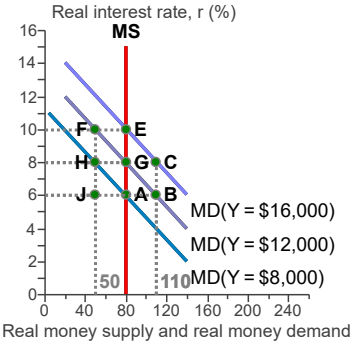
Why is the long-run aggregate supply curve vertical?

- **A.** because prices are fixed in the long run
- **B.** because the aggregate amount of output supplied is the full-employment level, regardless of the price level
- **C.** because the capital stock changes to produce the fixed amount of output in the long run
- **D.** because the price level depends on the amount of output

Answers: C. the *SRAS* curve is horizontal and the *LRAS* curve is vertical
A. because prices remain fixed in the short run
B. because the aggregate amount of output supplied is the full-employment level, regardless of the price level

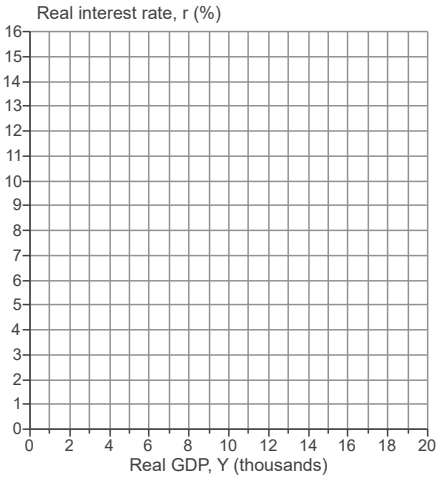
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30. Derive the *LM* curve graphically given the money market diagram.

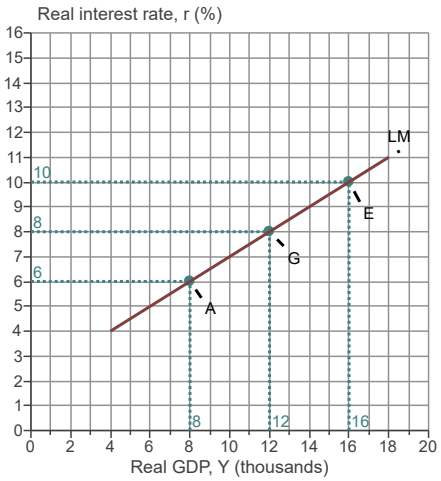


1. *Use the point drawing tool* to plot three points, one for each level of real GDP. Properly label each point.
2. *Use the line drawing tool* to connect these points and properly label this line.

Carefully follow the instructions above, and only draw the required objects.



Answer:



31. In an economy, the desired consumption and investment functions are given by

$C^d = 2{,}000 + 0.70Y - 4{,}500r$
$I^d = 2{,}000 - 4{,}000r$

where Y is output and r is the real interest rate.

Government purchases are $G = 2{,}100$.

The equation for desired national saving is:

$S_1^d = -4{,}100 + 0.30Y + 4{,}500r.$

Using the condition that at equilibrium, quantity of goods supplied = quantity of goods demanded, determine the real interest rates that clear the goods market when $Y = 10{,}000$ and $Y = 12{,}000$.

When $Y = 10{,}000$, the equilibrium rate of interest is:
$r =$ [ ] %.
(Enter your response as a percentage rounded to the nearest integer.)

When $Y = 12{,}000$, the equilibrium rate of interest is:
$r =$ [ ] %.
(Enter your response as a percentage rounded to the nearest integer.)

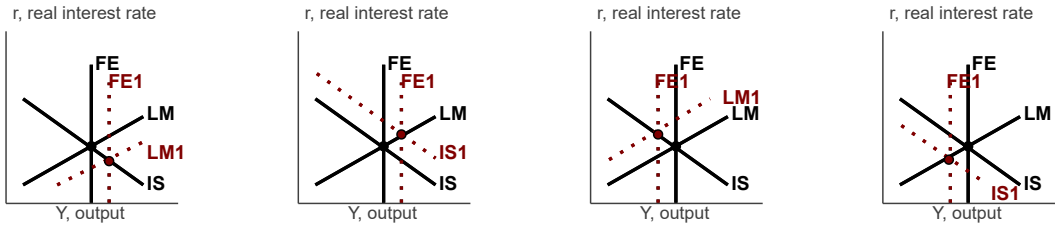
Answers: 36
29

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32. Suppose that the economy is in general equilibrium, and a temporary beneficial supply shock occurs which raises output above its previous full-employment level.

Which of the following graphs correctly shows the new equilibrium?

○ **A.** ○ **B.** ○ **C.** ○ **D.**



Answer:

A.

